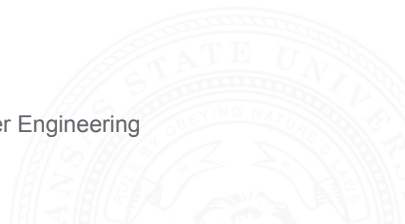


Line Parameters I

Build conductor resistance and inductance from material, geometry, and units

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Lecture outline

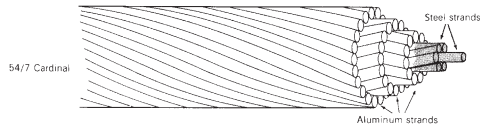
- 1 Part 1: Line configuration and units.
- 2 Part 2: Resistance.
- 3 Part 3: Inductance and reactance.
- 4 Part 4: Exercises.

Establish the physical model, calculate its parameters, then solve complete exercises.

PART 1

Line configuration and units

Overhead-line construction



A stranded conductor

Aluminum strands carry most of the current in this aluminum-conductor, steel-reinforced (ACSR) construction. The steel core provides mechanical support.

The metal area and outside radius describe different features of the conductor.

A circuit model connects this physical construction to voltage drop and power loss.

Source: Glover et al., sixth edition (2017), Fig. 4.1.

Series-line physics

Material and geometry

Conducting material, metal area, route length, and temperature determine the conductor's **resistance** R .

Resistance accounts for a resistive voltage drop and real-power loss.

Magnetic field and coupling

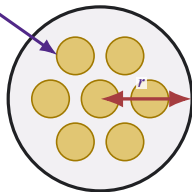
Current creates a magnetic field inside and around the conductors. Flux from one conductor also links circuits containing other conductors.

Inductance L relates flux linkage to current for a specified geometry. Changing current produces an inductive voltage drop.

Magnetic coupling makes flux linkage depend on the other currents and the return-path geometry.

Area and outer radius

A_{Al} sums the
conducting material



gray = envelope and gaps. All strands here are aluminum

Resistance input

A_{Al} : **effective conducting area.**
It sums the conducting metal across the strands. Gaps do not carry current.

Magnetic-model input

r : **outer radial field boundary.**
It marks the conductor surface where the external field begins.

Use the supplied aluminum area A_{Al} for resistance. The expression πr^2 applies when the conductor model defines a solid circular conductor.

Conductor quantities

Material and geometry

- ρ : resistivity ($\Omega \text{ m}$).
- A : conducting metal area (m^2).
- ℓ : conductor route length (m).
- r : physical outside radius (m).
- T : conductor temperature ($^{\circ}\text{C}$).
- R : conductor resistance (Ω).

Magnetic quantities

- I : conductor current (A).
- B : magnetic flux density (T).
- λ : flux linkage (Wb turn).
- L : inductance (H).

A prime will denote a quantity per unit length. Geometry and length must use compatible units.

Per-length parameters

For a uniform line, the same material and cross section repeat along the route.

Local parameter

R' is resistance per unit length, in Ω/km .

L' is inductance per unit length, in H/km or mH/km .

Route length

ℓ is the physical route length. A longer route contains more identical segments.

First determine the parameter of a unit length, then combine the segments.

A prime means per unit length. It does not mean a dimensionless per-unit (p.u.) value.

Compatible length units

The same physical rate can use different length units:

$$1 \text{ } \Omega/\text{km} = 10^{-3} \text{ } \Omega/\text{m} = 1.609344 \text{ } \Omega/\text{mi}$$

Geometry conversions

$$1 \text{ mm} = 10^{-3} \text{ m}$$

$$1 \text{ mm}^2 = 10^{-6} \text{ m}^2$$

Area conversion squares the length factor.

Route conversions

$$1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ mi} = 1.609344 \text{ km}$$

Use one consistent route unit.

Converting units changes the number, not the conductor or its physical properties.

PART 2

Resistance

Resistance per unit length

Uniform conducting material

For one unit of length, resistance depends on resistivity ρ and conducting area A :

$$R'_{\text{dc}} = \frac{\rho}{A}$$

Larger ρ increases R' . Larger metal area reduces R' .

Units and geometry

$$\frac{\Omega \text{ m}}{\text{m}^2} = \Omega/\text{m}$$

Use actual conducting metal area. For a solid circular conductor, $A = \pi r^2$; for a stranded record, use its supplied metal area.

Material and cross section determine the resistance rate before route length enters.

Source: Glover et al., sixth edition (2017), Sec. 4.2.

Temperature

Reference data

At reference temperature $T_0 = 20^\circ\text{C}$, ρ_{20} is resistivity and α_{20} is the resistance temperature coefficient ($^\circ\text{C}^{-1}$). k_T is the dimensionless temperature multiplier.

$$k_T = 1 + \alpha_{20}(T - 20^\circ\text{C}),$$

$$\rho_T = \rho_{20}k_T.$$

Same geometry

$$R'_{\text{dc},T} = \frac{\rho_T}{A} = R'_{\text{dc},20}k_T.$$

If $T > 20^\circ\text{C}$ and $\alpha_{20} > 0$, the resistance increases.

Keep the reference temperature attached to both ρ and α before applying the correction.

Resistance corrections

Apply the stated operating conditions to the rate

$$R'_{dc,20} = \frac{\rho_{20}}{A_{Al}}, \quad k_T = 1 + \alpha_{20}(T - 20^\circ\text{C})$$

$$R'_{ac,T} = R'_{dc,20} k_T k_{path} k_{ac,60}$$

- k_{path} : supplied correction for the strand current-path length.
- $k_{ac,60}$: supplied effective AC correction at 60 Hz.
- All multipliers are dimensionless and retain the declared conductor conditions.

Correct the per-length resistance first. These factors do not change the route length.

Total route resistance

Combine identical conductor segments

For a uniform route, resistance adds in series:

$$R = R'_{ac,T} \ell$$

A rate in metres

$$R [\Omega] = R' [\Omega/\text{m}] \ell [\text{m}]$$

Doubling ℓ doubles R but leaves R' unchanged.

A rate in kilometres

$$R [\Omega] = R' [\Omega/\text{km}] \ell [\text{km}]$$

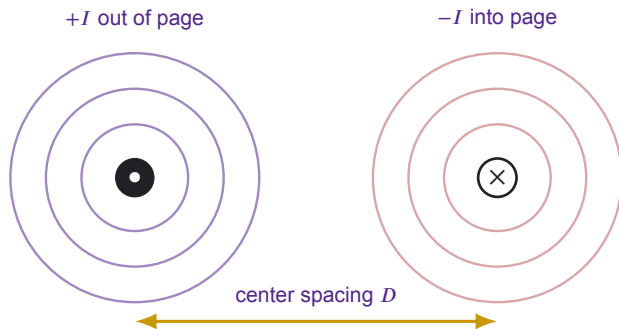
Use one rate and one compatible route length. Both unit choices describe the same total resistance.

PART 3

Inductance and reactance

Current path and magnetic field

The return conductor defines the linkage geometry



Magnetic field lines encircle each current.
Equal and opposite currents define a closed path. Their fields superpose and distant contributions cancel; the return wire is not a sharp field boundary.

$$\lambda = LI.$$

Inductance needs a closed current path. Return geometry is part of the result.

Flux linkage defines inductance

For fixed geometry and a linear magnetic model, current produces proportional flux linkage:

$$I \longrightarrow B \longrightarrow \lambda, \quad \lambda = LI.$$

Per unit of route length

$$\lambda' = L'I, \quad \boxed{L' = \frac{\lambda'}{I}}$$

The prime means per metre (or per kilometre), not per unit on a system base.

First find the magnetic linkage. No operating frequency is needed for the fixed, uniform-current model.

Physical meaning

L' measures linked magnetic flux per ampere and per unit of line length.

Specify the current distribution and return path before calculating it.

From inductance to AC reactance

The passive voltage drop is caused by *changing* flux linkage:

$$v_L(t) = \frac{d\lambda}{dt} = L \frac{di}{dt}.$$

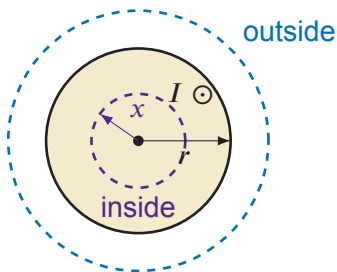
For $i(t) = I_m \cos \omega t$, differentiation shifts phase by $+90^\circ$:

$$\underline{V}_L = j\omega L \underline{I}, \quad \boxed{X_L = \omega L = 2\pi f L}.$$

geometry and magnetic model $\longrightarrow L \xrightarrow{\text{frequency } f} X_L$

Frequency enters when inductance becomes reactance. This separation assumes the stated current distribution; skin effect can change internal inductance.

Why both internal and external flux?



Cross section. Dashed circles:
magnetic-field paths.

Field exists in both regions

Inside: $0 < x < r$. Outside: $x > r$.

Current enclosed by a field circle is only part of I inside, but all of I outside.

$$\lambda' = \lambda'_{\text{int}} + \lambda'_{\text{ext}}$$

$$L' = L'_{\text{int}} + L'_{\text{ext}}$$

We are finding one conductor's contribution with a specified return path. The entire equal two-wire loop has twice this contribution.

Internal linkage: only part of the current

Assume a solid, nonmagnetic cylinder with uniform current density;
 $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$.

$$I_{\text{enc}} = I \frac{x^2}{r^2}, \quad H(x) 2\pi x = I_{\text{enc}}, \quad B(x) = \frac{\mu_0 I x}{2\pi r^2}.$$

For a one-metre axial strip of width dx , $d\Phi' = B(x) dx$. It links only the enclosed current fraction:

$$d\lambda'_{\text{int}} = \underbrace{\frac{x^2}{r^2}}_{\text{linked fraction}} B(x) dx.$$

$$L'_{\text{int}} = \frac{1}{I} \int_0^r d\lambda'_{\text{int}} = \frac{\mu_0}{2\pi r^4} \frac{r^4}{4} = \frac{\mu_0}{8\pi} = 0.05 \text{ mH/km}$$

The radius cancels only under these assumptions. Internal flux must be weighted by the fraction of current it links.

Source: Glover et al., sixth edition (2017), Sec. 4.4, Eqs. (4.4.1)–(4.4.10).

External linkage: why a logarithm?

Outside the conductor, a field circle encloses all of I :

$$B(x) = \frac{\mu_0 I}{2\pi x}, \quad d\lambda'_{\text{ext}} = B(x) dx.$$

With an equal return conductor at center spacing $D \gg r$, the assigned external contribution is approximately

$$L'_{\text{ext}} \approx \frac{1}{I} \int_r^D \frac{\mu_0 I}{2\pi x} dx = \boxed{2 \times 10^{-7} \ln \frac{D}{r} \text{ H/m}}.$$

- The logarithm comes from integrating the $1/x$ field.
- Return spacing sets the linkage reference; it is not a sharp field cutoff.

This is one conductor's contribution. Without a finite return-path specification, $D \rightarrow \infty$ gives no finite loop inductance.

Source: Glover et al., sixth edition (2017), Secs. 4.4–4.5.

GMR absorbs the internal contribution

Add the contributions assigned to one conductor:

$$L'_{\text{cond}} \approx 2 \times 10^{-7} \left[\ln \frac{D}{r} + \frac{1}{4} \right] = 2 \times 10^{-7} \ln \frac{D}{re^{-1/4}}.$$

Define the **geometric mean radius (GMR)** of this ideal solid conductor:

$$r' = re^{-1/4} = 0.7788r,$$

$$L'_{\text{cond}} \approx 2 \times 10^{-7} \ln \frac{D}{r'} \text{ H/m}.$$

- r' (also written D_s) is an effective radius. Here the prime does not mean per unit length.
- Real stranded conductors use their supplied GMR; $0.7788r$ is the ideal-solid result.

GMR absorbs internal flux linkage into an effective radius. The conductor has not physically become smaller.

Source: Glover et al., sixth edition (2017), Sec. 4.4, Eqs. (4.4.18)–(4.4.21).

Reactance and route totals

Convert the inductance rate at frequency f

$$\omega = 2\pi f,$$

$$X'_{\text{cond}} = \omega L'_{\text{cond}}$$

One conductor over the route

$$L_{\text{cond}} = L'_{\text{cond}} \ell$$

$$X_{\text{cond}} = X'_{\text{cond}} \ell = \omega L_{\text{cond}}$$

Equal outgoing and return conductors

$$Z_{\text{cond}} = R_{\text{cond}} + jX_{\text{cond}}$$

$$Z_{\text{loop}} = 2Z_{\text{cond}}$$

Frequency converts L' to X' . Length forms the total. Only the equal two-wire loop adds the factor of two.

PART 4

Exercises

Exercise 1: given and find

Given and find

For a uniform conductor, $\rho = 2.80 \times 10^{-8} \Omega\text{m}$, $A = 200 \text{ mm}^2$, and $\ell = 10.0 \text{ km}$. Ignore temperature and AC corrections.

- 1 Find resistance in Ω/km and the total resistance.
- 2 Predict the total resistance if the metal area doubles.
- 3 A student reports 0.00140Ω . Identify the missing conversion.

Exercise 1: solution

$$A = 200 \times 10^{-6} = 2.00 \times 10^{-4} \text{ m}^2$$

$$R' = \frac{\rho}{A} = 1.40 \times 10^{-4} \text{ } \Omega/\text{m} = 0.140 \text{ } \Omega/\text{km}$$

$$R = 0.140(10.0) = 1.40 \text{ } \Omega, \quad R_{2A} = 0.700 \text{ } \Omega$$

The 0.00140 result multiplies Ω/m by kilometres. Convert the rate or the length before multiplication.

Exercise 2: given

Fictional all-aluminum record + solid-cylinder magnetic surrogate

Fictional stranded all-Al, nonmagnetic record

Solid-cylinder magnetic surrogate

$$\begin{aligned}\ell &= 75.0 \text{ mi} = 120.7008 \text{ km}, \\ \rho_{20} &= 2.8264 \times 10^{-8} \text{ } \Omega\text{m}, \\ \alpha_{20} &= 0.00403/^{\circ}\text{C}, \\ A_{\text{Al}} &= 500 \text{ mm}^2, \quad T = 75^{\circ}\text{C}, \\ k_{\text{path}} &= 1.020, \quad k_{\text{ac},60} = 1.030\end{aligned}$$

$$r = 0.0140 \text{ m}, \quad D = 6.00 \text{ m},$$

$$f = 60 \text{ Hz}.$$

Uniform current is used for the internal-flux idealization. The return is an equal-conductor two-wire comparison.

A_{Al} and r are independent inputs. Use A_{Al} for resistance and r for the magnetic surrogate.

Exercise 2: find

Find and verify

- (a) Find the corrected resistance rate R' and one-conductor route resistance.
- (b) Find L'_{int} , L'_{ext} , total L'_{cond} , and X'_{cond} .
- (c) Calculate route L , X , and Z for one conductor, then the equal two-wire loop.
- (d) Verify the result using compatible metre and mile units.

State the units and model assumptions with each result.

Ex. 2(a): resistance rate

$$A_{Al} = 500 \text{ mm}^2 = 5.00 \times 10^{-4} \text{ m}^2,$$
$$R'_{dc,20} = \frac{2.8264 \times 10^{-8}}{5.00 \times 10^{-4}} = 5.6528 \times 10^{-5} \text{ } \Omega/\text{m},$$

$$R'_{dc,20} = 0.056528 \text{ } \Omega/\text{km} \approx 0.0909730 \text{ } \Omega/\text{mi}.$$

The units are consistent: ρ/A produces Ω/m , then the length-rate conversions preserve the same conductor.

Ex. 2(a): operating resistance

$$k_T = 1 + 0.00403(75 - 20) = 1.22165,$$
$$R'_{ac,75} = (5.6528 \times 10^{-5})(1.22165)(1.020)(1.030),$$

$$R'_{ac,75} \approx 7.25517 \times 10^{-5} \text{ } \Omega/\text{m} \approx 0.0725517 \text{ } \Omega/\text{km}.$$

The combined multiplier 1.28347 is specific to this conductor, temperature, and frequency.

Ex. 2(a): route resistance

Per-mile route

$$R'_{ac,75} \approx 0.0725517(1.60934) \\ \approx 0.1167607 \text{ } \Omega/\text{mi},$$

$$R_{\text{cond}} \approx 0.1167607(75) \approx 8.75705 \text{ } \Omega.$$

SI round trip

$$\ell = 75(1609.344) = 120700.8 \text{ m},$$

$$R_{\text{cond}} \approx (7.25517 \times 10^{-5}) \\ \times (120700.8) \\ \approx 8.75705 \text{ } \Omega.$$

The mile route and SI route agree. The result is one-conductor resistance, and the return model determines any loop total.

Ex. 2(b): magnetic components

Use r in the magnetic surrogate

$$\frac{D}{r} = \frac{6.00}{0.0140} \approx 428.571, \quad \ln(D/r) \approx 6.06046,$$

$$L'_{\text{int}} = \frac{\mu_0}{8\pi} = 5.00000 \times 10^{-8} \text{ H/m},$$

$$L'_{\text{ext}} \approx \frac{\mu_0}{2\pi} (6.06046) \approx 1.21209 \times 10^{-6} \text{ H/m}.$$

For this geometry, the external term is much larger. Both magnetic terms are independent of the resistance area A_{Al} .

Ex. 2(b): magnetic rates

Add the internal and external inductance rates

$$L'_{\text{cond}} \approx 0.0500000 + 1.21209 = 1.26209 \text{ mH/km}$$

At 60 Hz, $\omega = 2\pi(60) \approx 376.991 \text{ rad/s}$:

$$X'_{\text{cond}} = \omega L'_{\text{cond}} \approx 376.991(1.26209 \times 10^{-3}) = 0.475797 \text{ } \Omega/\text{km}$$

Equivalent rates in miles are

$$L'_{\text{cond}} \approx 0.00203114 \text{ H/mi}, \quad X'_{\text{cond}} \approx 0.765721 \text{ } \Omega/\text{mi}$$

These rates describe one conductor of the declared magnetic model. Route length has not yet entered.

Ex. 2(c): magnetic route totals

Inductance over 75 mi

$$\begin{aligned}L_{\text{cond}} &= L'_{\text{cond}} \ell \\&\approx 0.00203114(75) \\&\approx 0.152335 \text{ H}\end{aligned}$$

Check with total inductance:

Reactance over 75 mi

$$\begin{aligned}X_{\text{cond}} &= X'_{\text{cond}} \ell \\&\approx 0.765721(75) \\&\approx 57.4291 \Omega\end{aligned}$$

$$X_{\text{cond}} = \omega L_{\text{cond}} \approx 376.991(0.152335) = 57.4291 \Omega$$

The rate-times-length and total-inductance routes give the same reactance.

Ex. 2(c): conductor and loop

The factor of two has a narrow meaning

Composite teaching result: conductor

$$Z_{\text{cond}} \approx 8.757 + j57.429 \, \Omega.$$

$$R_{\text{cond}} = 8.757 \, \Omega, \quad L_{\text{cond}} = 0.152335 \, \text{H}.$$

Composite teaching result: loop

$$Z_{\text{loop}} \approx 17.514 + j114.858 \, \Omega.$$

$$R_{\text{loop}} = 2R_{\text{cond}}, \quad L_{\text{loop}} = 2L_{\text{cond}}.$$

The loop result applies to the displayed equal-conductor $+I/-I$ return model. A balanced three-phase series model instead uses conductor GMR and phase geometry.

Summary

- **Resistance:** material resistivity, conductor area, temperature, path length, and empirical AC factors determine series resistance.
- **Inductance:** $L' = \lambda'/I$; internal and external linkage add. GMR incorporates the internal term.
- **Reactance:** $X' = \omega L'$ converts the inductance rate to reactance. Multiply by ℓ once for the route total.
- **Model boundary:** conductor, loop, and phase quantities have distinct definitions. Declare the return path and geometry.

Preserve units and physical assumptions from the differential parameter to the total route impedance.